REGISTRATION CENTRE NUMBER	CENTRE NAME
CANDIDATE'S FULL NAMES	
CANDIDATE'S IDENTIFICATION NUMBER	SUBJECT CODE **0570** / PAPER NUMBER **2**

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GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

ORDINARY LEVEL

SUBJECT TITLE	SUBJECT CODE	PAPER NUMBER
MATHEMATICS	**0570**	**2**
	EXAMINATION DATE: **JUNE 2024**	

Duration: Two and a Half Hours

Enter the information required in the boxes above.
This paper is arranged in two sections, **A and B.**
Answer ALL questions in Sections A and B.

Section A: Answer ALL the questions in the spaces provided.
The mark allocation for each question is indicated.

Section B: All questions in Section B carry equal marks.

You are reminded of the necessity for good English and orderly presentation in your answers.
In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Calculators are allowed.

NUMBER	MARK
SECTION A	
SECTION B	
1	
2	
3	
4	
TOTAL	

FOR EXAMINERS' USE ONLY

Marked by: ----------

Signature:---------- *Date* ----------

Checked by: ----------

Signature:---------- *Date* ----------

SCORE

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SECTION A

ANSWER ALL 15 QUESTIONS IN THIS SECTION

1. Evaluate $\left(2\frac{1}{2}-1\frac{1}{5}\right)\div\frac{3}{5}$.

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(4 marks)

2. Simplify $3(x-2y)-2(x+5y)$.

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(4 marks)

3. Given that $f=\frac{uv}{u-v}$, express u in terms of f and v.

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.....

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(5 marks)

4. Given the sets $Q=\{x;\ x \text{ is a factor of } 18\}$ and $R=\{x:\ x \text{ is a multiple of } 3 \text{ less than } 18\}$.
List the elements of set:

a) Q .....

b) R .....

c) Find $Q\cap R$ .....

d) Draw a Venn diagram to represent the two sets.

.....

.....

.....

(7 marks)

5. Complete the truth table in Figure 1.

p	q	$\sim q$	$p \wedge \sim q$
T	T		
T	F		
F	T		
F	F		

Figure 1

(6 marks)

6. Given the matrices $A = \begin{pmatrix} 3 & x+1 \\ 2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & 14 \\ 4 & y-x \end{pmatrix}$.

a) Find 2A.

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.....

.....

b) Given that 2A = B. Find the values of x and y.

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.....

(7 marks)

7. The table shows values of x and its corresponding values of y

x	3	b	9
y	30	15	10

a) State whether the relationship between x and y is direct or indirect variation.

.....

Find:

b) the constant of proportionality

.....

.....

.....

.....

c) the value of b

.....

.....

.....

(6 marks)

8. Given the network in Figure 2.

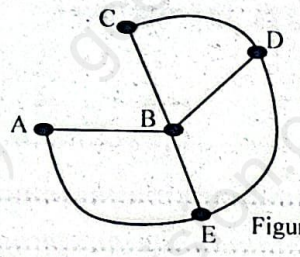


Figure 2

State the number of :

a) regions (R).....

b) vertices (V).....

c) edges (E) .....

d) Hence , show that R + V - 2 = E.

.....

.....

(6 marks)

9. Given the points O, A and B with coordinates (0, 0), (5, 0) and (1, 4) respectively.
Determine as column vectors:

a) $\overrightarrow{OB}$,

.....

.....

b) $\overrightarrow{OB} - \overrightarrow{OA}$,

.....

.....

c) $\overrightarrow{OM}$, where M is the midpoint of $\overrightarrow{AB}$.

.....

.....

(6 marks)

10. Solve the pair of simultaneous equations

$x - 2y = 5$

$3x + y = 8$

.....

.....

.....

.....

(5 marks)

11. Figure 3 shows a circle with chords AC and BD intersecting at the centre P. Given that angle DPC = 86^0, AP = 6cm, PC = 12cm and BP = 8cm.

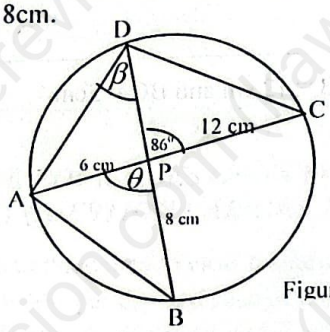


Figure 3

Find the:

a) angle θ

.....

.....

b) angle β

.....

.....

c) length of PD.

.....

.....

(6 marks)

12 Figure 4 shows a sector of a circle of radius 10cm. Given that the length PQ is 12cm and the area of the sector is 64.4 cm^2.

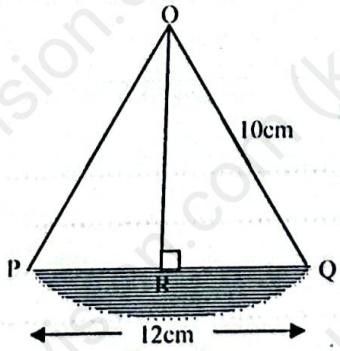


Figure 4

Find the:

a) length RQ

.....

.....

b) length OR

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.....

c) area of the shaded region.

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(7 marks)

13. Given triangle ABC in figure 5 with AB = 12 cm and BC = 5cm.

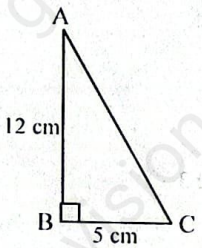


Figure 5

Find:

a) the length AC

.....

.....

b) angle ACB, to the nearest degree.

.....

.....

.....

.....

(6 marks)

14. Solve the inequality $3x - 5 > x - 1$

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.....

(4 marks)

15. The probability that a student passes in the first sequence examination is $\frac{3}{5}$ and passes in the second sequence examination is $\frac{9}{10}$. Find the probability that the student:

a) fails the first sequence,

.....

.....

b) passes in both sequences.

.....

.....

(4 marks)

SECTION B

ANSWER ALL FOUR QUESTIONS IN THIS SECTION. EACH QUESTION CARRIES 15 MARKS

1. A farmer employed two students and three men to clear his farm for two weeks, working 5 days a week. The two students each received 2,000 FCFA per day and each of the three men received 3,000 FCFA per day.
   a) Calculate the total amount that the farmer spent on his workers for clearing his farm within two weeks.
   Given that the total amount the farmer used to buy chemicals for spraying is $\frac{3}{5}$ of the total amount spent on clearing.
   b) Calculate the amount used for buying chemicals. Other expenses summed up to 35,000 FCFA.
   c) Calculate his total expenses.
   Given also that he sold all his produce at 620,500FCFA and 30% of his profit was shared between his first and the second wives in the ratio 3:2.
   d) Find the amount shared between his two wives.
   e) Express the amount received by the first wife as a percentage of the amount shared between his wives.

(15 marks)

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2. (i) The table shows the distribution of marks scored by 100 candidates in an examination.

Mark (x)	60 - 62	63 - 65	66 - 68	69 - 71	72 - 74
Frequency (f)	5	18	42	27	8

State the:

a) number of classes or intervals in the distribution,
b) lower class limit of class 63 – 65,
c) upper class limit of class 72 – 74 ,
d) modal class of the distribution ,
e) class mark of the class 60 – 62,
f) class size of the distribution .

(10 marks)

(ii) The functions f and g are defined on $\Re$, the set of real numbers by $f : x \mapsto 2x + 3$ and $g : x \mapsto x - 3$

Find:

a) $f(2)$
b) $gof(x)$

(5 marks)

3. (i) On a graph paper and taking 1 cm for 1 unit on both axes.

a) Draw triangle PQR with coordinates P(-1, 4), Q(- 1, 1) and R(- 4, 1).

b) Find the coordinates of the image $P'Q'R'$ of triangle PQR under the transformation matrix

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

c) Draw triangle $P'Q'R'$ and describe the transformation completely.

(8 marks)

(ii) Using a pencil, a ruler and a pair of compasses only.

a) Construct triangle ABC such that AB = AC = BC = 7 cm .

b) Bisect any two angles in triangle ABC and label the point O, where the bisectors meet.

c) With O as centre, draw a circle such that AB, AC and BC are tangents to the circle.

d) Measure and write down the length of the diameter of the circle.

(7 marks)

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4. (i) Given the function $f(x) = 3x - x^2$ defined for $-1 \le x \le 4$.

a) Construct a table of values of $y = f(x)$ for integral values of x in the range.

b) Using a scale of 2 cm to represent 1unit on both axes, draw the graph of $y = f(x)$.

From your graph,

c) solve the equation $3x - x^2 = 0$,

d) state the maximum value of $f(x)$.

(9 marks)

(ii) Given the polynomial $p(x) = (x-2)(2x-1)(x+1)$.

a) Solve the equation $p(x) = 0$.

b) Find the remainder when $p(x)$ is divided by $(x-1)$.

(6 marks)

STOP

GO BACK AND CHECK YOUR WORK